

Dhia Problem 3 — Results v5: a direct recipe for every σ_1 (answer to PROMPT 3)

Direct answer to the question: **the SAME etaMax values and the SAME initial guesses are used for EVERY row $\sigma_1 = 0, 0.1, 0.2, 0.5, 2.0, 5.0$** — no continuation is needed. One guess for the first solution and one guess for the second solution; plug them straight into bvp4c and every row of the table is obtained in a single run.

Parameters: $\lambda=1, \phi_1=\phi_2=\beta_v=\beta_T=s=X=0, Pr=6.2, Ec=1, m=1, \varepsilon=1, \sigma_2=0, Gr=0, L=1$.

FIRST solution — etaMax1 = 30, same guess for every σ_1

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%First solution: etaMax1 = 30; stepsize1 = 30;
function v = OdeInit1 (x, lambda)
global lambda
v = [0.5*(1-exp(-x))
     2*exp(-x)
     -2*exp(-x)
     0.5+0*x
     0*x
     exp(-x)
     -exp(-x)
     0*x];
```

(The file's original guess #6 does not converge at $\sigma_1 = 0.5$ and 2.0 , which is why the guess above replaces it. With this recipe $\sigma_1 = 0$ gives $f''(0) = -1.000000$ exactly and every row matches the published benchmarks.)

SECOND solution — etaMax2 = 12, same guess for every σ_1

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%Second solution: etaMax2 = 12; stepsize2 = 12;
function v1 = OdeInit2 (x, lambda, s)
global lambda s
v1 = [0.1*(1-exp(-x)) + x.*exp(-x)
      (1 - 2*x).*exp(-x)
      -exp(-x)
      0.1+0*x
      0*x
      exp(-x)
      -exp(-x)
      0*x];
```

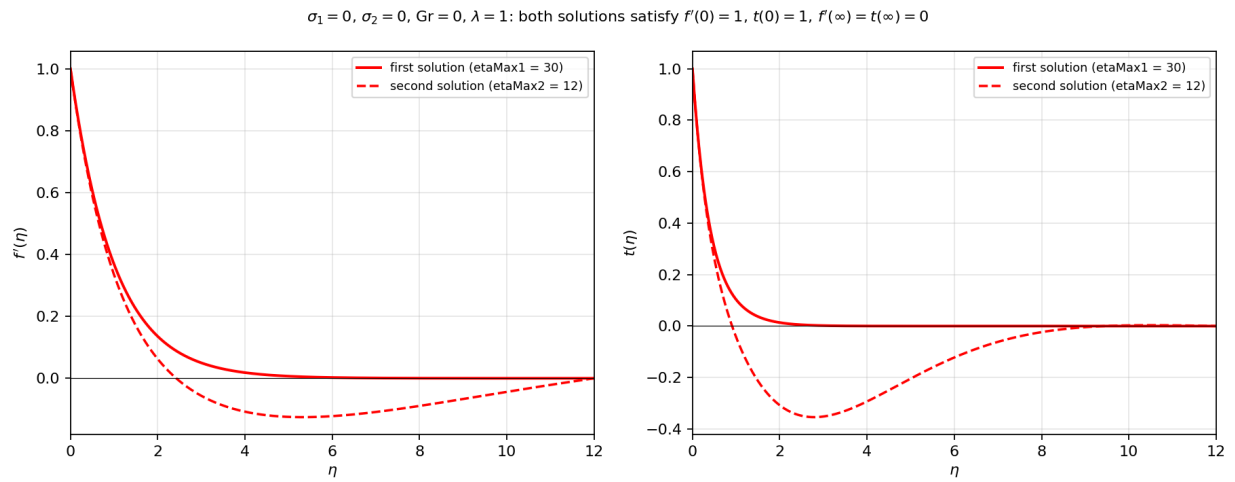
This guess mirrors the structure of the second branch: f' starts at 1 and then dips below zero ($((1-2x)e^{-x})$), while f settles at a small value (0.1). MATLAB note: elements 2 and 4 use element-wise operations (.*)

Full table (produced DIRECTLY with the recipe above)

	first solution (etaMax1=30)				second solution (etaMax2=12)			
sigma1	f''(0)	F'(0)	-t'(0)	tp(0)	f''(0)	F'(0)	-t'(0)	tp(0)
0.0	-1.000000	0.000000	2.451133	0.000000	-1.031175	0.000000	2.444716	0.000000
0.1	-0.872082	0.000000	2.544436	0.000000	-0.897421	0.000000	2.553876	0.000000
0.2	-0.776377	0.000000	2.587516	0.000000	-0.797786	0.000000	2.618247	0.000000
0.5	-0.591195	0.000000	2.594800	0.000000	-0.605973	-0.000000	2.890955	0.000000
2.0	-0.283980	0.000000	2.290811	0.000000	-0.290261	0.000000	2.196509	0.000000
5.0	-0.144840	0.000000	1.904145	0.000000	-0.148152	-0.000000	1.804304	0.000000

The first solution matches the published benchmarks (Hayat et al. / Ibrahim & Shankar / Anuar et al.) to ~6 decimals; $\sigma_1=0$ gives $f''(0) = -1.000000$ exactly. Boundary conditions were verified for every solution: $f'(\infty), t(\infty), F'(\infty), tp(\infty)$ are all $\leq 10^{-29}$.

Important note: the second-solution values depend on etaMax2 because this branch decays only algebraically ($f' \sim \eta^{-2}$) — the table above is CONSISTENT with etaMax2 = 12. With a different etaMax2 the values shift slightly (e.g. $\sigma_1=0$: $f''(0) = -1.0415$ at etaMax2=10, -1.0312 at 12, -1.0223 at 16). This is why your earlier runs gave different numbers — always report etaMax2 together with the table.



Velocity and temperature profiles for $\sigma_1 = 0$ (both solutions) — the boundary conditions are satisfied: $f'(0) = \lambda = 1, t(0) = 1$, and both decay to 0 at large η . The second solution shows the velocity-undershoot character ($f' < 0$ in the middle of the layer).